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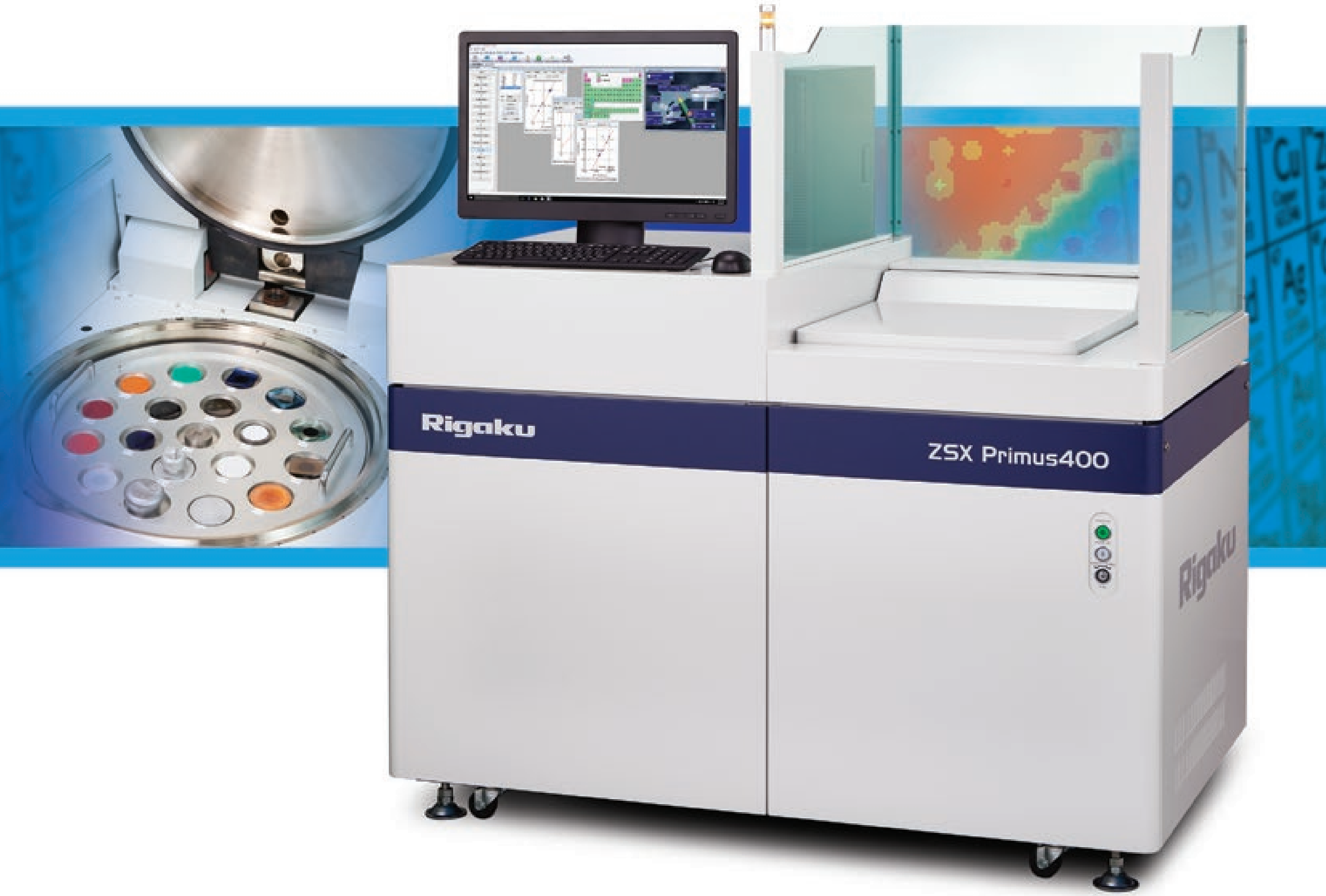
ZSX Primus 400

Sequential wavelength dispersive X-ray fluorescence spectrometer



Elemental analysis of large objects by WDXRF spectroscopy, with micro-mapping

Rigaku
POWERING NEW PERSPECTIVES

ZSX Primus 400

with micro-mapping and large object capabilities

Fast, precise elemental analysis of beryllium (Be) through uranium (U)

The Rigaku ZSX® Primus 400 sequential wavelength dispersive X-ray fluorescence spectrometer provides unequaled flexibility, sensitivity and reliability – the three core requirements that underlie the superior value inherent in all Rigaku XRF instrumentation. ZSX Primus 400 features the new ZSX Guidancen software with the built-in XRF know-how of a skilled spectroscopist. Since the introduction of our first sequential WDXRF – the Rigaku GF-S – in 1962, we have incorporated this essential philosophy into all aspects of our instrument design, development and manufacture. Through continued focus on scientific innovation and ongoing partnerships with the world's leading research institutions and industrial laboratories, Rigaku has changed the face of XRF analysis and broadened the scope of XRF applications deployed throughout the world.

Flexibility

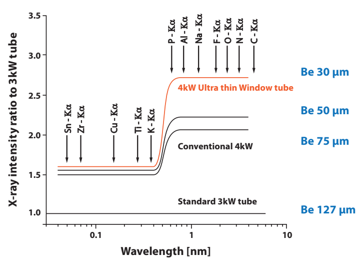
ZSX Primus 400 is available with a variety of adapters for samples of various sizes as well as with irregular shapes. Samples can be up to 400 mm in diameter and 50 mm high with a maximum weight of 30 kg.

Reliability

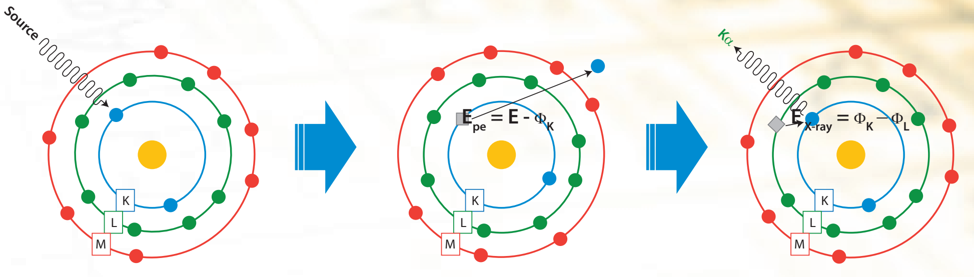
Rigaku XRF spectrometers are the most reliable in the world. ZSX Primus 400 is available with automatic pressure control for superior light element performance, and the sample chamber is designed for easy cleaning. Automatic cleaning of the proportional counter detector center wire ensures consistent performance.

Sensitivity

At the heart of an XRF spectrometer is the X-ray tube. The available Super-Trace™ 30 4.0 kW end window X-ray tube delivers unparalleled performance.

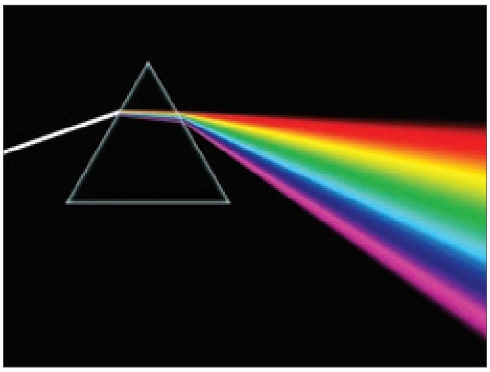


What is XRF?



X-ray fluorescence schematic

In X-ray fluorescence (XRF), an electron can be ejected from its atomic orbital by the absorption of X-rays (photons) from an X-ray tube. When an inner orbital electron is ejected (middle image), a higher energy electron transfers to fill the vacancy. During this transition, a *characteristic* photon may be emitted (right image) that is of a unique energy for each type of atom. The number of *characteristic* photons per unit time (counts per second or cps) is proportional to the amount of that element in a sample. Thus, qualitative and quantitative elemental analysis is achieved by determining the energy of X-ray peaks in a sample spectrum and measuring their associated count rates.



Analyzing crystals disperse radiation in the X-ray spectral region in the same way that a prism spreads the spectrum of visible light

How WDXRF works

Wavelength dispersive X-ray spectroscopy (WDXRF) is a method used to separate and measure the *characteristic* fluorescent X-rays emitted from a sample. The technique employs an analyzing crystal to spatially spread the X-ray light, much like a prism spreads visible light into its component colors. The wavelength of the impinging X-ray and the crystal's lattice spacings are related by Bragg's law and produce constructive interference when they satisfy the Bragg equation. The X-rays emitted by the sample are collimated by parallel metal blades (Soller slit) and irradiate an automatically selected analyzing crystal at a precise angle. X-ray light diffracted by the analyzing crystal is spatially spread out, so that *characteristic* photons may be collected by a detector positioned at a precise angle to record the X-ray intensity of a specific element.

X-ray optics

In WDXRF, the sample, analyzing crystal and detector are all mounted on a variable angle optical mechanism called a goniometer. As illustrated (at right), the goniometer precisely varies the angles so that the detector can record X-ray fluorescence from any element of interest. The entire X-ray optical mechanism is contained within a vacuum chamber to eliminate absorption of soft radiation (low-energy photons) by the air.

The ZSX Primus 400 spectrometer can support up to ten analyzing crystals that can be automatically selected depending on the wavelength being analyzed, enabling elements from almost the entire periodic table to be analyzed.

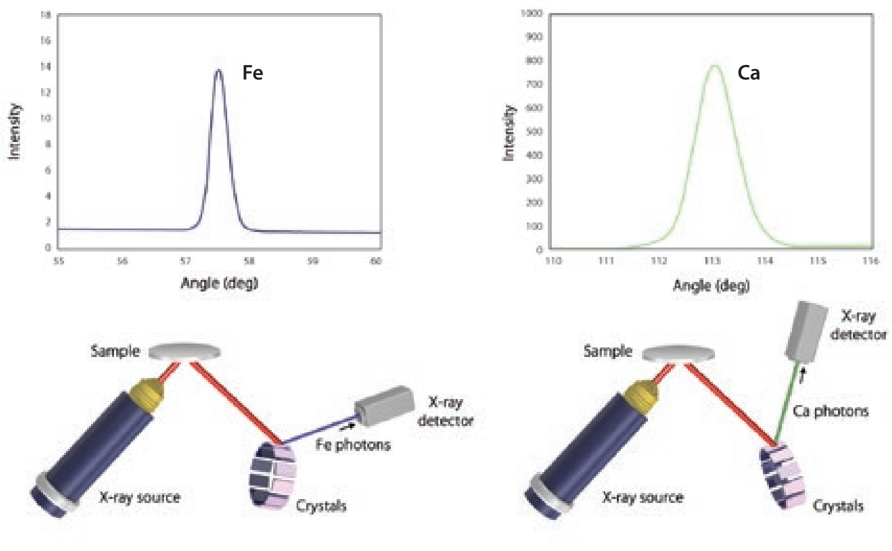
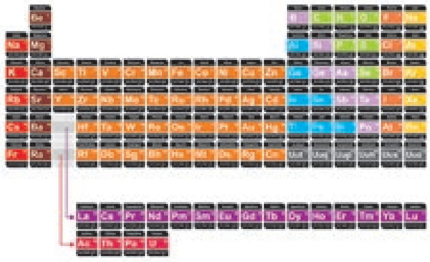


Illustration of how different analyzing crystals and angles allow various elements to be measured

Exceptional flexibility for large and heavy samples

ZSX Primus 400



Analyze Be through U

Broad elemental analysis coverage for maximum analytical flexibility.

Largest laboratory XRF sample chamber

Sample chamber can house a sample of up to 400 mm in diameter, up to 50 mm in height and 30 kg in mass.

ZSX Guidance software

ZSX Primus 400 features the new ZSX Guidance software with the built-in XRF know-how of a skilled spectroscopist.

r-θ stage for elemental mapping

Provides information about sample uniformity and surface contamination. A 400 mm circular or 180 x 250 mm rectangular sample may be fully mapped.

Vacuum system is standard

For improved light element sensitivity and suppression of the argon (Ar) peak.

Synthetic multi layers, RX-SERIES*

The new synthetic multi layer crystal "RX85" produces about 30% greater intensity than existing multi layers for B-Kα.

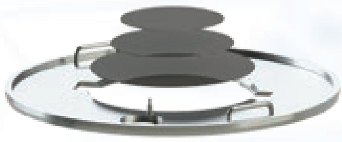
Highly sensitive curved crystals

Curved PET and Ge crystals are incorporated in the standard configurations. The intensity for P and S by curved Ge increases by 30% compared with flat Ge. The intensity for Al and Si by curved PET increases by 30% compared with flat PET. Sensitivities in semi-quantitative analysis by SQX analysis could be enhanced with the curved crystals.

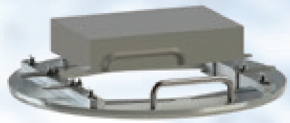
* Rigaku has been developing proprietary synthetic multi-layers with innovated technology.

Standard XRF samples

Magnetic disks



5,6,8,12 inch wafer



Rectangular or irregularly shaped samples

Variety of available sample adapters

Optional adapters are available for wafers and disks of various sizes and other samples with irregular shapes.

Available safety system for the optics

A safety system consisting of protective film (optional) in front of the collimator and a beryllium filter (optional) in front of the X-ray tube prevents serious damage caused by sample breakage during measurement.

Variable spot size analysis

Quantification of small regions of a sample down to 500 µm.

Flow-bar based graphical user interface (GUI)

Provides guidance through the calibration process and random access to interface windows. Flow-bar represents a "best practice" in analytical software design.

30 µm Be end-window X-ray tube

SuperTrace 30, the thinnest end-window tube available in the industry, provides unsurpassed light element (low-Z) sensitivity and faster low-Z element analysis times.

Available camera for sample observation

Arbitrary locations of a sample may be measured while observing the sample image (20 x 20 mm field-of-view).

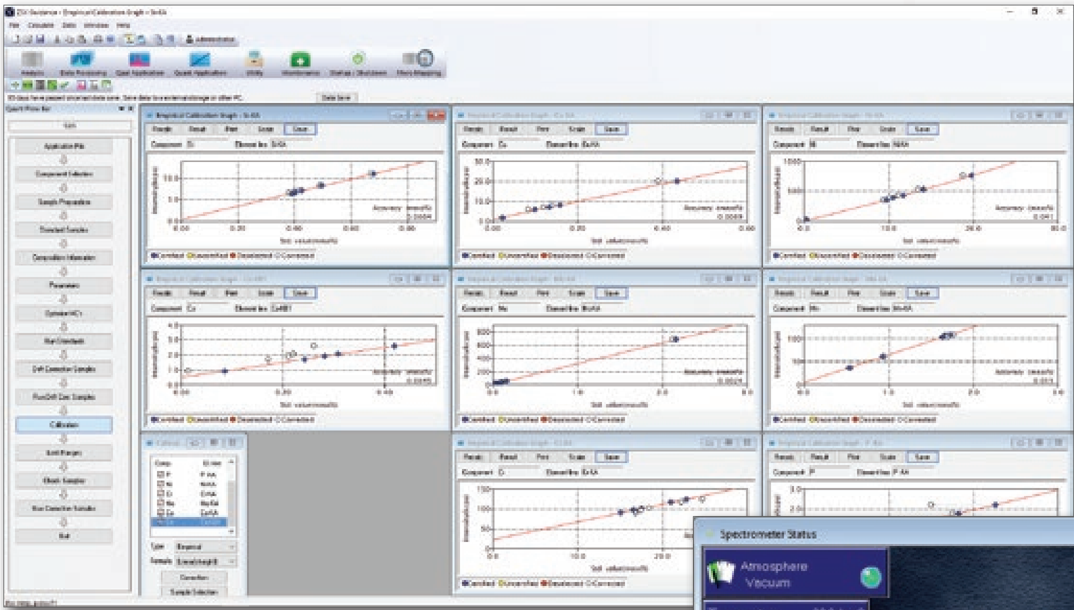
Automatic center wire cleaning

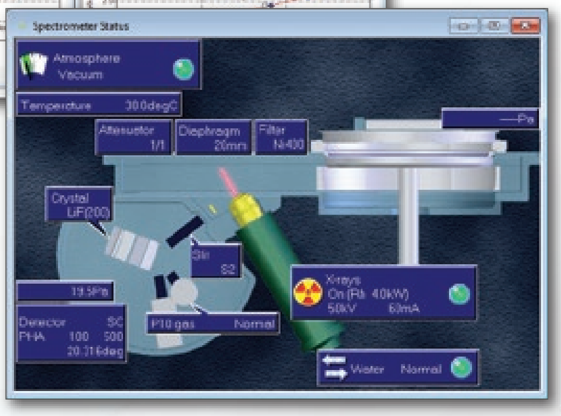
Flow proportional counter detector (F-PC) features automatic center wire cleaning to reduce routine maintenance.

Graphical user interface

Powerful software that is easy to use

With an eye toward the future, Rigaku has combined extensive experience in applications development and unsurpassed technical knowledge to create the world's best XRF analytical software. With a firm belief that knowledge is power, Rigaku has developed software that is not only user-friendly, but sophisticated and powerful enough for the most complex analysis. The ZSX Primus 400 Windows® based software was conceived and built with end-user needs and requirements in mind.





Spectrometer status window

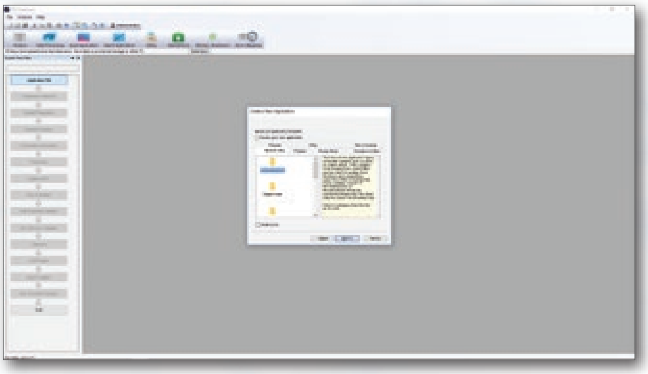
Graphical status display

Another famous Rigaku innovation is the graphical instrument status window. Real time display of key parameters allows users to assess the condition of the spectrometer at a glance. This feature saves the operator time and increases situational awareness.

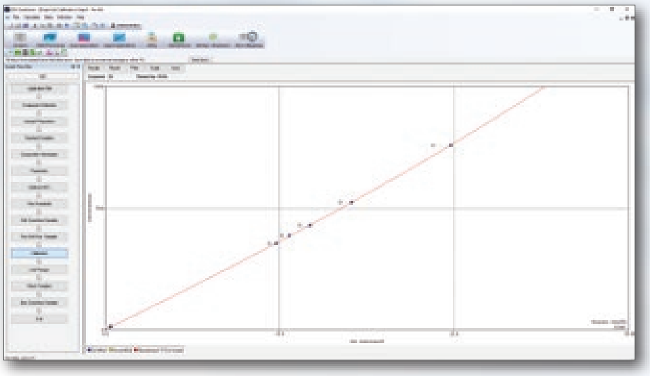
Flowbar guidance makes it easy

Based on the famous Rigaku easy-to-use flowbar interface, ZSX Primus 400 software walks the user through the steps required to set up either an empirical or a fundamental parameters application. For empirical calibrations, the flowbar covers every detail, from setup of an application file to the selection of a template and the components to be measured. The user is then guided through acquisition parameters setup, the setup of standards and drift correction, through calibration and reporting.

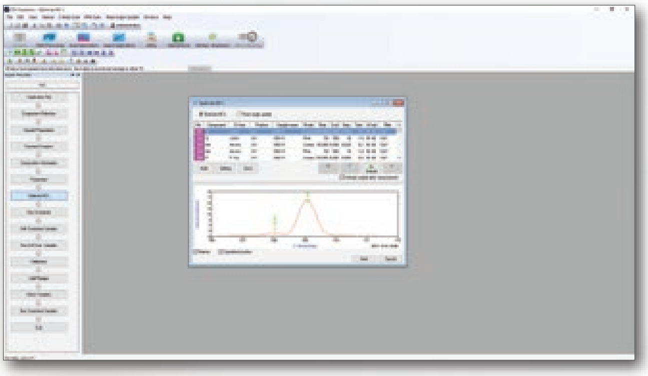
Flowbar



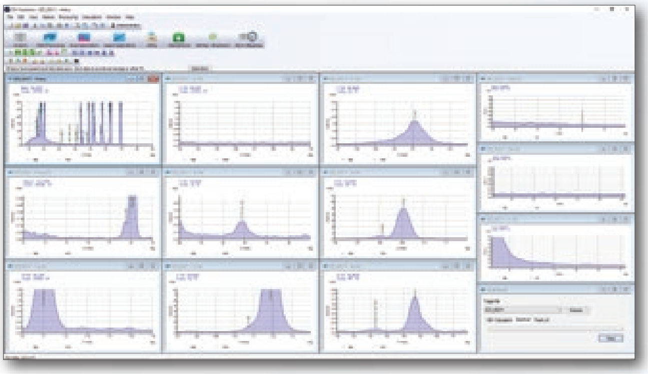
Templates for qualitative and quantitative applications guide you through each setup



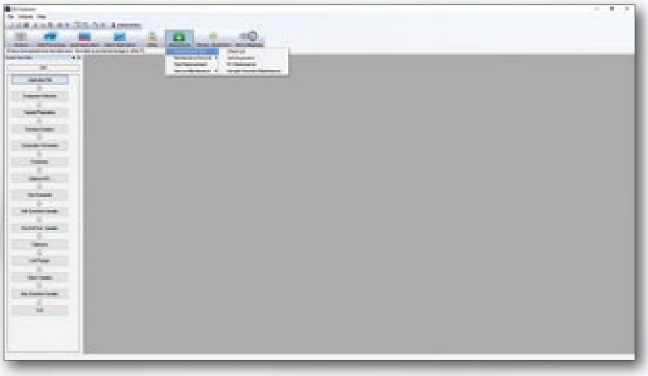
Calibration windows are clear and easy to understand



The "Optimize" window allows the adjustment of a variety of conditions, ensuring the best response possible



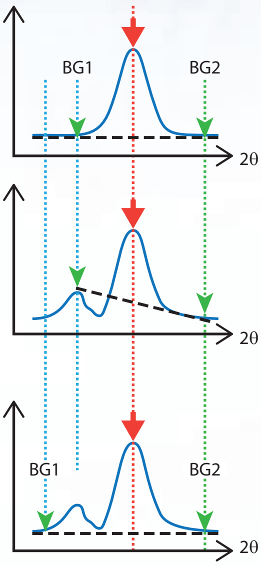
Qualitative scans are organized for easy interpretation and automatically list potential interfering elements



Built-in diagnostics maximize uptime and reliability

SQX fundamental parameters

Rigaku ZSX Primus 400 spectrometer is available with Rigaku SQX fundamental parameters (FP) software that provides semi-quantitative analysis of almost all sample types without standards – and rigorous quantitative analysis with standards. SQX greatly reduces the number of required standards, for a given level of calibration fit, as compared to conventional empirical methods. As standards are expensive and can be difficult to obtain for many applications, SQX can significantly lower the cost of ownership and reduce workload requirements for routine operation.



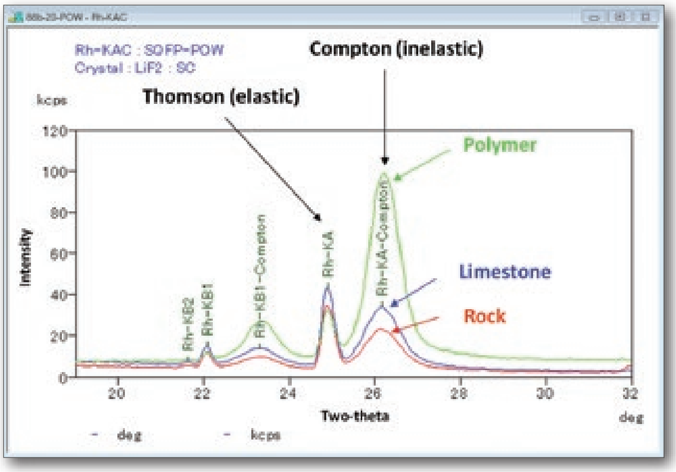
FA example

Fixed Angle (FA) analysis

Unlike other software using fixed angles (top), SQX features automatic background point selection to eliminate errors in net peak intensity (middle). Patented FA analysis (bottom) delivers higher measurement precision and lower detection limits due to dramatically improved counting statistics.

Scattering FP method

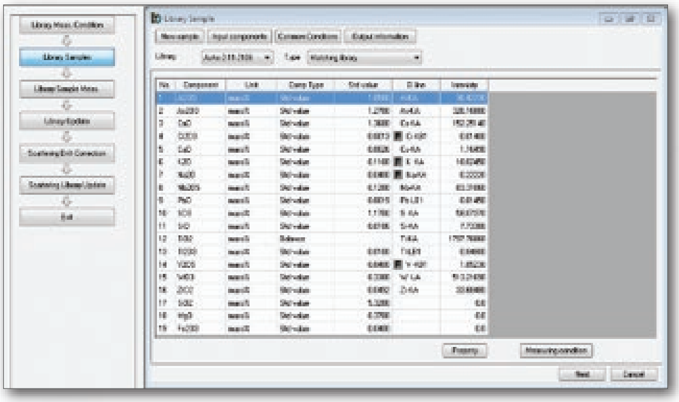
Accurate analysis, without analyzing major light element (low-Z) components, is now possible. Previously, it was difficult to accurately analyze samples – such as fly ash, soil, tissue or polymers – unless all the invisible elemental components were defined properly. With the newly developed Scatter FP method, scatter lines are used to estimate the effect of these non-analyzed components. Scattering FP employs a computational model, using Monte Carlo simulations, that considers the distribution of scattering angles for each geometrical location of the sample. It also considers the effect of photo-electron excitation for ultra-light elements.



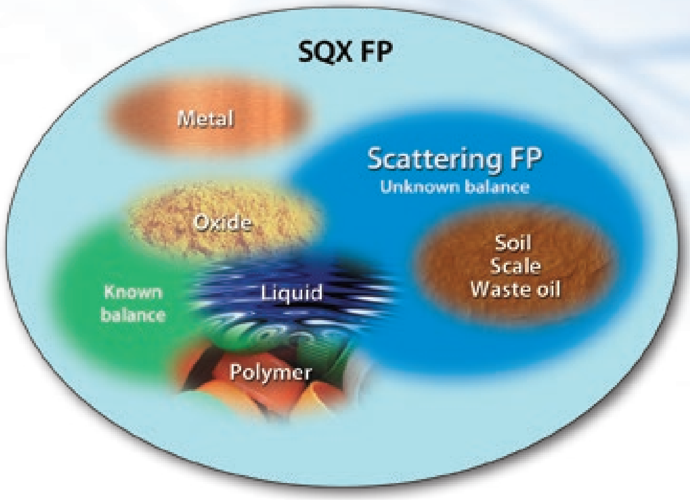
Thompson/Compton ratio by sample type

Coal ash by SQX Scattering FP method (units: mass%)

Element	Na	Mg	Al	Si	P	S	K	Ca	Ti
Reference value	1.69	1.27	12.6	23.6	0.26	0.196	1.05	6.36	0.79
Scattering FP	2.14	1.08	12.6	20.3	0.25	0.28	1.04	6.23	0.79
FP (Balance: C)	1.2	0.59	6.74	10.7	0.13	0.14	0.534	3.19	0.4
Element	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga
Reference value	0.0243	0.0075	0.05	3.64	0.0037	0.0032	0.0122	0.0063	(0.0045)
Scattering FP	0.025	0.0112	0.0515	3.42	0.0049	0.0055	0.013	0.0064	0.0044
FP (Balance: C)	0.013	0.0057	0.0258	1.72	0.0025	0.0028	0.0066	0.0033	0.0022
Element	As	Rb	Sr	Y	Zr	Nb	Pb		
Reference value	0.0029	0.0054	0.11	(0.0054)	(0.046)	(0.0025)	0.0047		
Scattering FP	0.0035	0.0052	0.113	0.0046	0.031	0.0015	0.0038		
FP (Balance: C)	0.0017	0.0027	0.0586	0.0024	0.017	0.0008	0.0019		



Scan using Matching Libraries



Matching libraries

SQX Matching Libraries can be created to match your sample type. By integrating a few standards of a similar matrix to unknown sample types, and creating a semi-quantitative library for each, far greater accuracy can be realized for the unknowns. Analysts can create as many matching libraries as needed to provide turn-key, accurate semi-quantitative results for their materials. With SQX Matching Libraries, semi-quantitative analysis by FP becomes truly quantitative.

SQX FP features and benefits

- **Matching Libraries turns semiquantitative analysis into quantitative analysis**

  Use Rigaku's Matching Libraries or build a custom library with your standards

- **Scattering FP and photo-electron correction for superior matrix modeling**

  Reliably accounts for effects of the unanalyzed low-Z element portion of the sample

- **Fundamental parameters based line overlap correction for complex spectra**

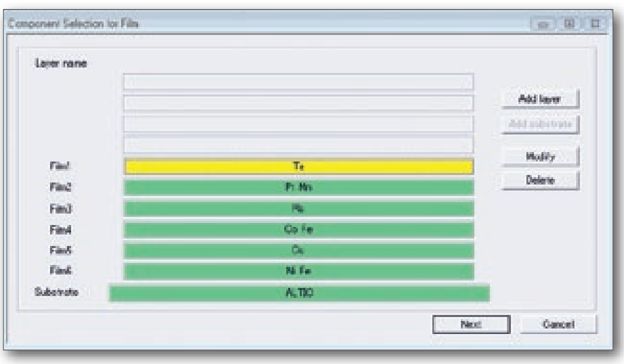
  Overlap factors conveniently adapted to both bulk and thin film samples using theoretical intensities

Quant fundamental parameters

Rigaku quantitative FP software has a variety of unique functions for applications like thin films, geological samples, and fused beads.

Coatings and multi-layer thin films

Thin film materials play an important role in many of today's core industries. Applications to determine thin film composition and thickness are vital in the production of semiconductor devices, industrial coatings, and solar energy conversion devices. XRF measurements offer precise, non-destructive elemental analysis capabilities that can be used across the broad spectrum of application requirements.



Quant FP features and benefits

- **Advanced matrix correction works synergistically with Compton scatter**

  Enhanced accuracy for heavy elements in rocks, ores and other geological samples

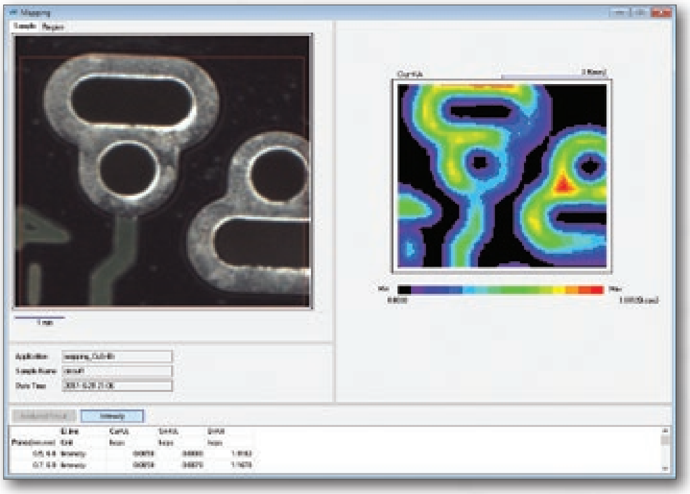
- **Matrix, dilution, and evaporation corrections for fused beads**

  Mathematically account for flux or sample LOI without pre-ignition of samples

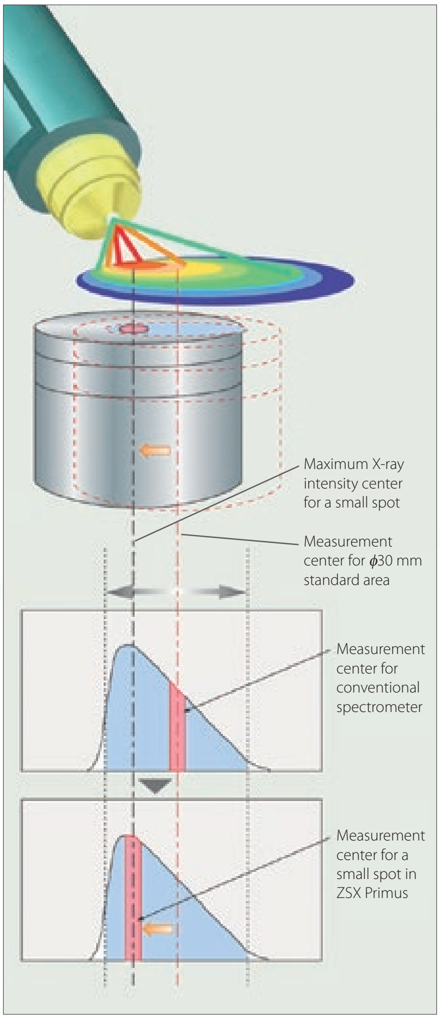
Micro-spot, multi-point analysis and large area mapping

Elemental mapping for anisotropic samples

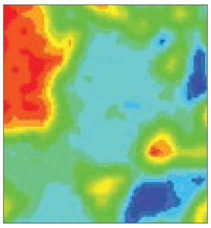
ZSX Primus 400 spectrometer is configured with micro-mapping capabilities, allowing the user to measure areas as small as 500 microns on the surface of a sample. The ability to elementally map a sample surface (100 µm mapping resolution) can provide valuable information about sample uniformity, allow quantification of small regions of a sample, and examine surface contamination. An automated (r-θ) sample stage, a camera system, and intelligent software position the sample so the brightest portion of the X-ray beam is centered on the small spot.



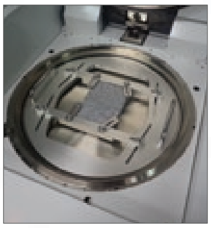
2D map showing copper (Cu) distribution on printed circuit board



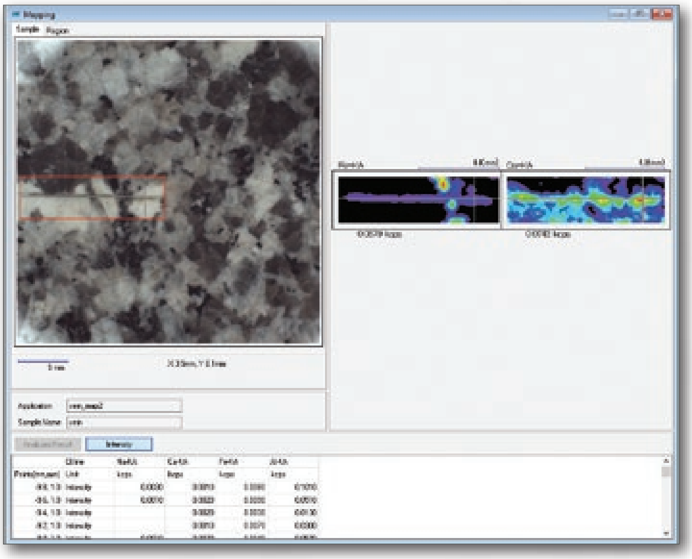
The r-θ positioning stage ensures that the "hot spot" of the X-ray beam is aligned with the region of interest, vastly improving the sensitivity for small areas



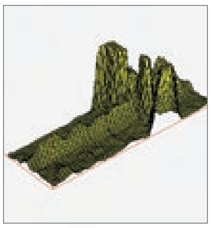
2D silicon (Si) micro-mapping result for granite rock



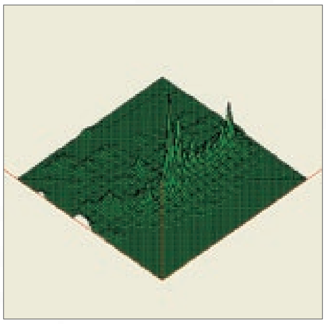
Large sample by using multi-attachment



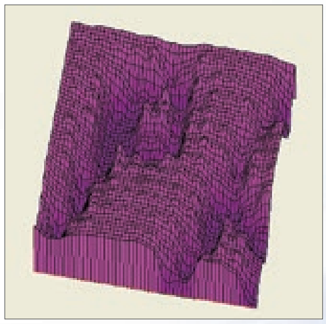
Camera view and mapping results of iron (Fe) and calcium (Ca) in small vein cutting of granite rock



3D map of iron (Fe) in granite rock vein



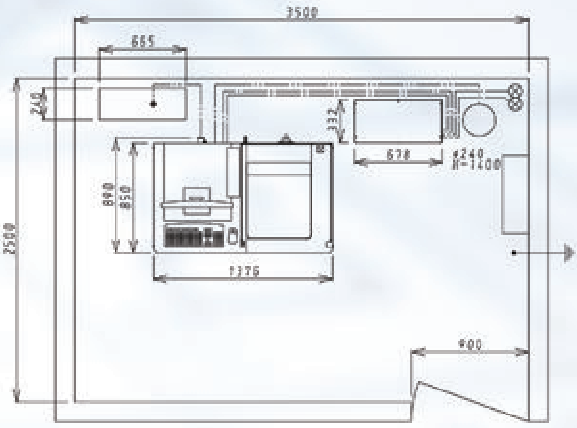
3D map revealing of sulfur-rich zone in laminated sedimentary rock

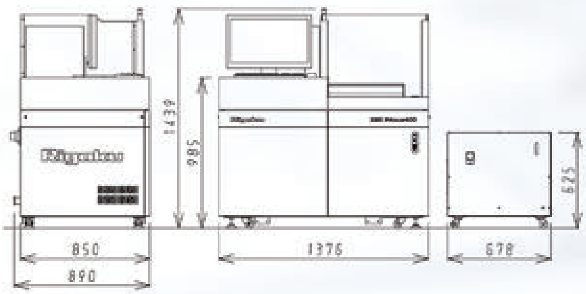


3D phosphorus map of scratched region of zinc phosphate rust proof coating

Specifications and installation requirements

Specifications		
Outline	Elemental range	$_{4}$Be – $_{92}$U
	Optics	Wavelength dispersive
X-ray generator	X-ray tube	End window type, Rh-anode
		3 kW or 4 kW
	High voltage generator	High frequency inverter system
		Max. Rating: 4 kW, 60 kV – 150 mA
Spectrometer	Maximum sample size	400 mm diameter, 50 mm height, ≤30 kg. Optional adapters are available for various size samples with irregular shapes.
	Primary X-ray filters	Automatic exchanger (Max. 7 filter) Ni400, Ni40, Al125, Al25 (standard) Ti, Sn (Diffraction interference rejection, optional) Be30 (Protection of X-ray tube, optional)
	Analysis area diaphragm	ϕ 30, 20, 10, 1, 0.5 mm
	Crystal exchanger	Maximum 10 crystals
Measurement	Detector	Heavy elements: SC Light elements: F-PC





Installation requirements	
Power	3 phase 200 V 50 A, Single phase 100-240 V 50/60 Hz (PC)
Ground	Independent ground with resistance less than 30 Ω
Cooling water	Quality: Equivalent to drinking water Temperature: Lower than 30 °C Pressure: 0.29-0.49 MPa Flow: More than 5 L/min
Drainage	Open drainage
Room temperature	15-28 °C with daily variation within ±2°C
Humidity	Lower than 75% RH
Vibration	Lower than human sensitive level
P-10 gas	Ar 90% Methane 10% Mixture gas pressure 0.15 MPa

Backed by Rigaku

Since its inception in 1951, Rigaku has been at the forefront of analytical and industrial instrumentation technology. Today, with hundreds of major innovations to our credit, the Rigaku Group of Companies are world leaders in the field of analytical X-ray instrumentation. Rigaku employs over 1,400 people worldwide in operations based in Japan, the U.S., Europe, South America and China.

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POWERING NEW PERSPECTIVES

ZSX Primus 400

Sequential wavelength dispersive
X-ray fluorescence spectrometer

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